



University of Rochester

Laboratory XV **The Franck-Hertz Experiment**

DEPARTMENT OF PHYSICS & ASTRONOMY
PHYSICS 123 - 183
WAVES AND MODERN PHYSICS

Objective

Accelerate electrons through argon vapor in a vacuum tube and measure the current at the anode as a function of accelerating voltage, reproducing the landmark 1914 experiment of Franck and Hertz. Identify the periodic current drops caused by inelastic collisions to determine the energy of the first excited state of argon and confirm the discrete nature of atomic energy levels.

Theory

The Franck-Hertz experiment, performed by James Franck and Gustav Hertz in 1914, is the first direct demonstration of discrete electron energy levels in an atom.

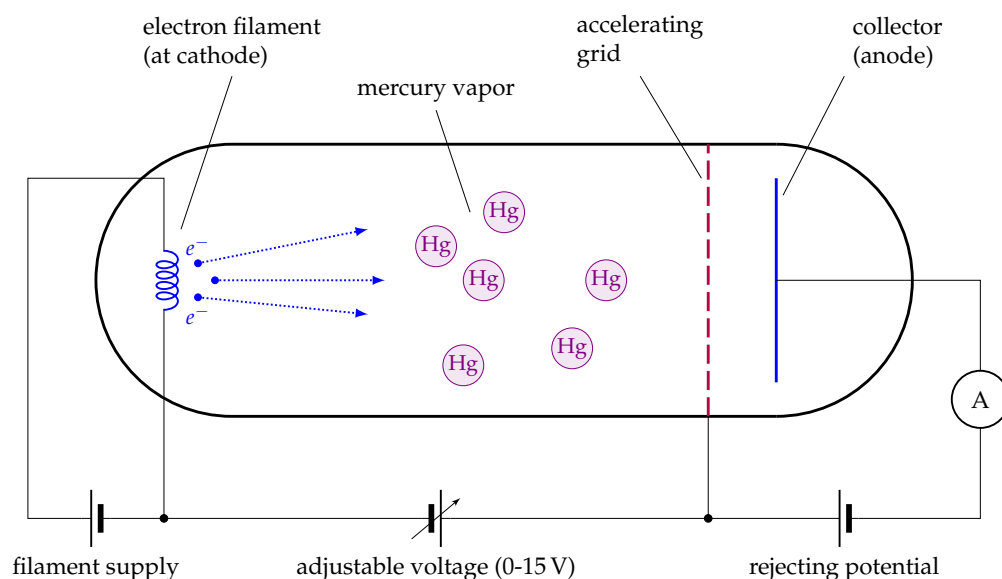


Figure 1: A schematic diagram of the original Franck-Hertz vacuum tube.

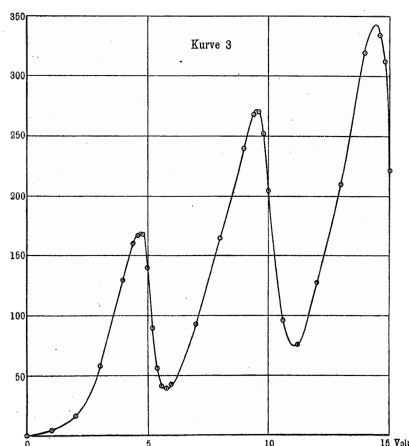


Figure 2: Anode current in the vacuum tube (y-axis) versus cathode-grid voltage (x-axis). Credit: J. Franck and G. Hertz, *Verh. Dtsch. Phys. Ges.* 16:457, 1914.

Franck and Hertz studied energetic electrons traveling through a glass vacuum tube (Fig. 1) filled with a small amount of gaseous mercury (Hg). The electrons were emitted from a hot electrode inside the tube next to a low-voltage contact called the **cathode**. A metal grid held at an adjustable voltage relative to the cathode caused the electrons to accelerate toward the grid. On the far side of the grid, another electrode called the **anode** collected the electrons, allowing Franck and Hertz to measure the current of the electron beam after it passed through the Hg vapor.

As they increased the adjustable voltage, Franck and Hertz found the anode current increased proportionally with the voltage until 4.9 V, at which point the anode current dropped close to zero. As they continued increasing the voltage beyond 4.9 V, the current again increased proportionally until dropping close to zero at 9.8 V (exactly twice 4.9 V). This pattern repeats in steps of 4.9 V. You can see the pattern in Fig. 2, which shows the anode current versus adjustable grid voltage published by Franck and Hertz in 1914.

In a follow-up experiment, Franck and Hertz noticed a blue-violet glow in the Hg gas in the tube, peaking almost entirely at

$\lambda = 254 \text{ nm}$. This wavelength corresponds to a photon energy of

$$E_{\text{photon}} = \frac{hc}{\lambda} = \frac{(4.15 \times 10^{-15} \text{ eV s})(2.998 \times 10^{17} \text{ nm/s})}{(254 \text{ nm})} \approx 4.9 \text{ eV}.$$

Note that 4.9 eV is the kinetic energy of an electron accelerated from rest by a potential difference of 4.9 V, the same as the period observed in Fig. 2. This is not a coincidence. When the accelerated electrons collide with Hg atoms in the gas, at most energies they elastically scatter off the atoms, changing direction but losing no energy. However, at energies of 4.9 eV, the electrons collide *inelastically* with the Hg atoms, losing their energy to atomic electrons, which then re-emit the absorbed energy as 254 nm photons. The process is shown schematically in Fig. 3.

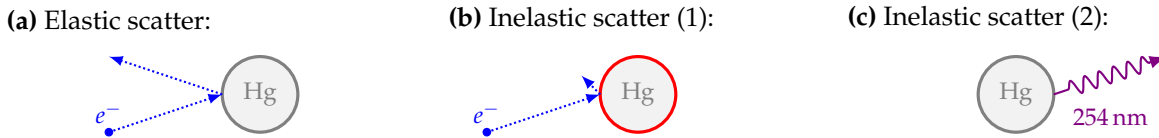


Figure 3: In elastic scattering of an accelerated electron from an Hg atom (a), the electron changes direction but loses no energy. In inelastic scattering, the accelerated electron loses 4.9 eV to a valence electron in the Hg atom, promoting it from the ground state to the first excited state (b). The valence electron then de-excites and emits a 254 nm photon (c).

In 1913, Niels Bohr published his atomic model of quantized electron energy levels. The Bohr model successfully describes the discrete emission and absorption spectra of hydrogen-like atoms¹, explaining the spectra as transitions of atomic electrons between quantized energy levels. The Franck-Hertz experiment is consistent² with this picture:

- Accelerated electrons in the Franck-Hertz tube collide elastically with Hg atoms at most voltages.
- At 4.9 V and multiples of 4.9 V, a beam electron inelastically transfers its kinetic energy to a valence electron in atomic mercury, exciting it from the atom's ground state, $\text{Hg}(6^1\text{S}_0)$, to the first excited state, $\text{Hg}(6^3\text{P}_1)$. These two states differ in energy by 4.9 eV.
- The excited electron then de-excites back to the ground state, causing the emission of a 254 nm photon.

For his theoretical work on atomic structure, Niels Bohr received the Nobel Prize in Physics in 1922. James Franck and Gustav Hertz were awarded the Nobel Prize in 1925 for their experimental verification of quantized atomic structure.

Experiment

You will repeat the Franck-Hertz measurement in this lab, but instead of using Hg, we will use a tetrode (a vacuum tube with four penetrating electrodes) filled with argon (Ar) gas. Switching to Ar does not change the nature of the experiment, but it avoids the use of Hg, a neurotoxin that must be heated to become a vapor (Hg is a liquid at room temperature).

A schematic view of the setup is shown in Fig. 4. In contrast to the original Franck-Hertz device, our setup includes two accelerating grids with separate voltage adjustments. The four electrodes on the device give you control of four settings:

1. The potential at the cathode K, which will be connected to ground.
2. The voltage between the cathode and the first accelerating grid, $V_{G1,K} = 1.5 \text{ V}$.
3. The voltage between the cathode and the second accelerating grid, $V_{G2,K} = 0 \text{ V to } 100 \text{ V}$.

¹Atoms with one electron in the valence shell.

²Note that the Bohr model does not exactly describe the Hg line spectrum, since mercury is not a hydrogenic atom.

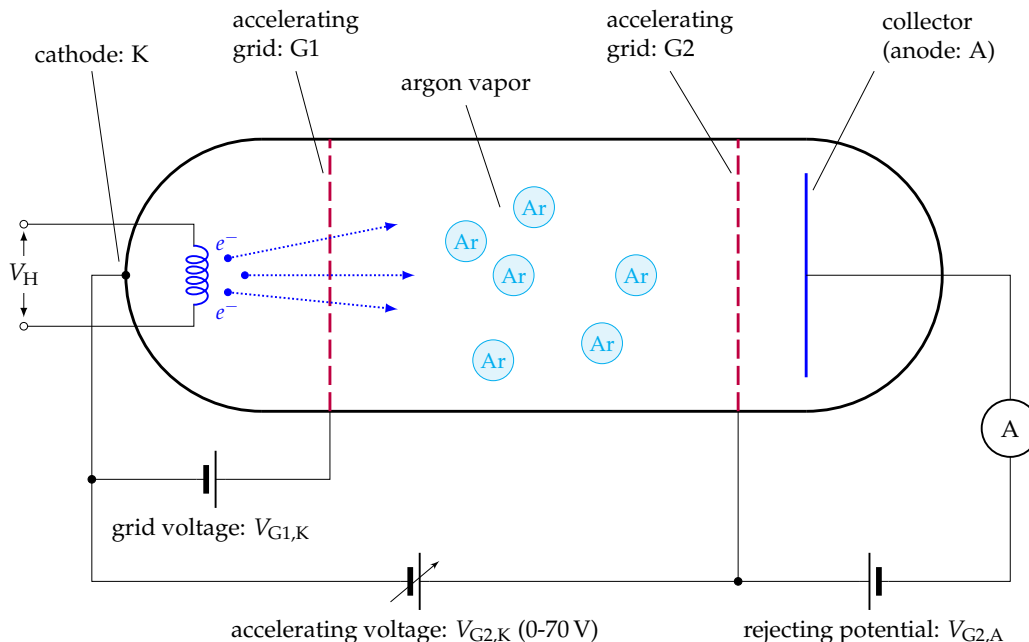


Figure 4: The tetrode filled with Ar gas used in this lab.

4. A small rejecting potential $V_{G2,A} < 10 \text{ V}$ between the second grid and the anode.

By measuring the anode current I_A with an ammeter, you will characterize the electron beam as a function of the adjustable voltage $V_{G2,K}$. You can also change the filament voltage, V_H , to alter the intensity of the electron beam.

Data Collection for the Experiment

During the experiment, you will start with a low value of $V_{G2,K}$. The accelerated electrons in the tube will collide elastically with the Ar gas, so you will see I_A increasing proportionally with $V_{G2,K}$.

When $V_{G2,K}$ reaches the first excitation potential³ of Ar, the electrons colliding with Ar atoms near grid G2 will interact inelastically with the argon, transferring their kinetic energy to the atoms' valence electrons. The beam electrons will then have too little energy to overcome the reverse field created by the rejecting potential $V_{G2,A}$, causing I_A to drop significantly.

With a further increase of $V_{G2,K}$, the beam electrons will be **re-accelerated** by the larger grid voltage after their first inelastic collision with the Ar atoms. The re-accelerated electrons will obtain enough kinetic energy to overcome the reverse field created by $V_{G2,A}$, allowing them to reach the anode and causing I_A to rise proportionally with $V_{G2,K}$.

Eventually, $V_{G2,K}$ will be twice the excitation potential of Ar, and the electrons re-accelerated to grid G2 will inelastically collide a second time with Ar atoms, losing their energy and causing I_A to drop again. You can observe this pattern repeating every time $V_{G2,K}$ reaches an integer multiple of the excitation potential. An example scope trace

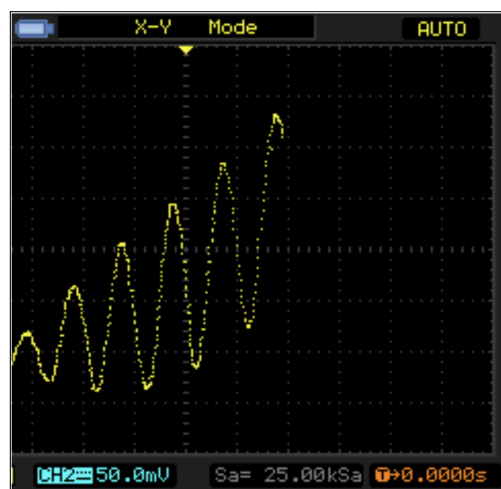


Figure 5: Example oscilloscope trace of anode current I_A versus grid voltage $V_{G2,K}$ for Ar.

³Experimentally, you'll observe the Ar excitation potential somewhere between 10 V and 13 V.

is shown in Fig. 5. By measuring the spacing between dips, we determine the first excitation potential of Ar and verify that the energy absorbed by the atom is discrete.

Use of the Equipment

WARNINGS: Read the following directions before using the equipment.

1. During the experiment, pay close attention to the output current indicator when the adjustable voltage $V_{G2,K} > 60$ V. This is especially important at the higher voltage (filament voltage) and sensitivity (current multiple) settings. If the current meter reading increases suddenly, decrease the voltage at once to avoid damage to the Franck-Hertz tube.
2. If you want to change the value of $V_{G1,K}$, $V_{G2,A}$, and V_H during the experiment, **first rotate the “0-100 V” voltage adjustment knob counter-clockwise all the way to the end** to turn off the grid voltage before making your changes.
3. The filament voltage of this instrument is $V_H = 3$ V, 3.5 V, 4 V, 5 V, 5.5 V, and 6.3 V. You can experiment with these filament voltages, but beware that too high a filament voltage can cause current distortion. In such a condition, V_H should be decreased.

Procedure

Adjust the Operating Voltages

1. Make yourself familiar with the control knobs and read the warning carefully before starting the experiment. Due to the risk of damage to the tube, the use of filament voltage $V_H > 5$ V **is discouraged** except under the direct supervision of a TA/TI.
2. Make sure the setup has a vacuum tube installed (ask the TA if unsure) and covered prior to use.
3. The 0-100 V knob is continuously adjustable with a soft stop. Use care to **slowly rotate** this knob during use and when zeroing. Turn all voltage adjustment knobs fully counterclockwise (make all zero). Turn the “Manual-Auto” switch to “Manual” and rotate the “Scan Knob” counterclockwise all the way to the end.
4. **Set the filament voltage $V_H = 4$ V and the “Current Multiple” selector to 10^{-8} .** Turn on the device: the green power indicator will glow. **Preheat the bulb for 2 minutes before starting your experiment.**

Warning: NEVER change the “Current Multiple” selector with the instrument powered on.

When turning the unit off, the current meter will dip and hold under zero momentarily while the device is powering down.

5. Set $V_{G1,K} = 1.5$ V:
Move the “Voltage Stepper” switch to 1.3–5 V, and rotate the 1.3–5 V adjustment knob until the voltmeter reads 1.5 V.
6. Set $V_{G2,A} = 7.5$ V:
Move the “Voltage Stepper” switch to 1.3–15 V, and rotate the 1.3–15 V adjustment knob until the voltmeter reads 7.5 V (rejecting voltage).

Operation in Manual Mode (OPTIONAL)

Skip to the next section unless you are explicitly asked by the TA to run in manual mode.

Once you have completed the voltage setup described above, move the “Voltage Stepper” switch to 0-100 V and rotate the 0-100 V adjustment knob counterclockwise until the voltmeter reads 0 V (soft zero). The total accelerating voltage is then equal to the rejecting potential $V_{G2,A} = 7.5$ V, which sets the starting point for the sweep. Remember that this knob is continuously adjustable with a soft zero. You can now gradually increase $V_{G2,K}$ and record measurements of I_A with the ammeter and voltmeter.

Operation in AUTO Mode (DEFAULT)

1. Complete the voltage adjustment steps described above.
2. Turn on the oscilloscope. Press the **Menu** button and set it to **XY** mode by using buttons on the right side of the display.
3. Set up the marker channel using the **Cursor** button. Once configured, the X and Y marker positions can be adjusted using the **Off** button.
4. Connect the oscilloscope to the Franck-Hertz kit as shown in Fig. 6. Connect the instrument X output to channel 1 of the oscilloscope and the Y output to channel 2. Also connect the instrument's ground to the oscilloscope ground.



Figure 6: Oscilloscope probe connection to the voltage panel of the Franck-Hertz kit.

5. Move the **Voltage Stepper** on the Franck-Hertz kit to 0–100 V and the operation switch to **Auto**. If you have connected everything properly, the signal should look as shown in Fig. 7.

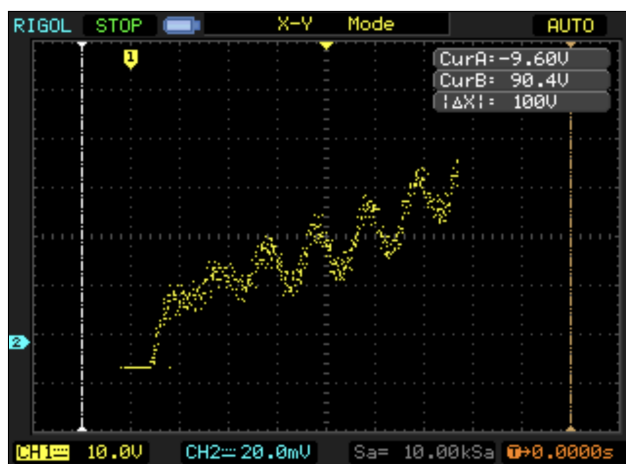


Figure 7: Oscilloscope display of an auto-scanned signal from the Franck-Hertz kit.

6. Measure the peak-to-peak voltage differences using the oscilloscope cursors and fill in Data Table 1 in the Postlab exercises.

7. You can also save your data in a flash drive in a different format. For that, use the **Acquire** button on the top panel.